
Answer all questions. Calculators and mobile phones are NOT allowed.

1. [10 pts.] Given that $A^{-1} = \begin{bmatrix} 3 & 5 & 1 \\ 1 & 2 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

(a) Find $\text{adj}A$.

(b) Find $A \text{adj}A$.

2. [10 pts.] Solve $\begin{vmatrix} 1 & 0 & 0 \\ 2 & x & 1 \\ 1 & 3 & x-5 \end{vmatrix} = \begin{vmatrix} x & -1 \\ 2 & x \end{vmatrix}$, for x .

3. [10 pts.] Let $U = (1, 1, 2, -3, 1)$. Find all scalars k such that $\|kU\| = 4$.

4. [10 pts.] Let X and Y be vectors in $\mathbb{R}^n$ such that $\|X\| = \|Y\| = 2$. Show that:

$$\|X + Y\|^2 + \|X - Y\|^2 = 16.$$

5. [10 pts.] Let A be an $m \times n$ matrix. Show that the solution set of the homogenous linear system $AX = O$ in n unknowns is a subspace of $\mathbb{R}^n$.

6. [10 pts.] Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $X = (x_1, x_2)$ and $Y = (y_1, y_2)$ in V and $k \in \mathbb{R}$:

$$X + Y = (x_1 + y_1 - 2, x_2 + y_2 - 2), \quad kX = (kx_1, kx_2).$$

Show that V is not a vector space under the given operations.

7. [10 pts.] Let $U_1 = (1, 2, 2, -1)$, $U_2 = (4, 9, 9, -4)$ be vectors in $\mathbb{R}^4$. Determine whether the vector $U = (5, 8, 9, -5)$ is in $\text{span}\{U_1, U_2\}$.

8. [10 pts.] Determine whether the vectors $U_1 = (1, 2, 0)$, $U_2 = (2, 3, 0)$, and $U_3 = (4, 3, 1)$ span $\mathbb{R}^3$.

9. [$5 \times 4 = 20$ pts.] In each of the following questions, select the correct answer.

I. If $X = (1, 2, 1)$ and $Y = (5, 1, 0)$ are vectors in $\mathbb{R}^3$, then $\|2X - Y\|$ is equal to:

(A) $\sqrt{22}$.

(B) $\sqrt{10}$.

(C) $2\sqrt{6} + \sqrt{26}$.

(D) $2\sqrt{6} - \sqrt{26}$.

II. Which one of the following sets is a subspace of $\mathbb{R}^2$:

- (A) $W = \{(a, a + b) : a, b \in \mathbb{R}\}.$
- (B) $W = \{(a, b) \in \mathbb{R}^2 : a + b = 1\}.$
- (C) $W = \{(a, b) \in \mathbb{R}^2 : a^2 + b^2 = 16\}.$
- (D) $W = \{(a, b) \in \mathbb{R}^2 : |a + b| > 1\}.$

III. Which one of the following sets is linearly independent in $\mathbb{R}^3$:

- (A) $\{(1, 2, 4), (-1, -2, -4)\}.$
- (B) $\{(1, 2, -1), (1, 0, 1), (0, 0, 0)\}.$
- (C) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (4, 5, 7)\}.$
- (D) $\{(1, 2, -1), (1, 0, 0), (1, 2, 0)\}.$

IV. Let V be the set of positive real numbers, and consider the following addition and scalar multiplication operations on $X = x$ and $Y = y$ in V and $k \in \mathbb{R}$: $X + Y = xy$ and $kX = x^k$. Then, the negative of a vector $X = x \in V$ is:

- (A) $-X = \frac{1}{x}.$
- (B) $-X = -x.$
- (C) $-X = x.$
- (D) $-X = 1.$

1. [10 pts.] Given that $A^{-1} = \begin{bmatrix} 3 & 5 & 1 \\ 1 & 2 & 0 \\ 4 & 0 & 1 \end{bmatrix}$.

(a) Find $\text{adj}A$.

We have $\text{adj}A = |A|A^{-1}$. Since $|A^{-1}| = -7$, therefore $|A| = -\frac{1}{7}$.

Thus, $\text{adj}A = -\frac{1}{7}A^{-1}$.

(b) Find $A \text{adj}A$.

$A \text{adj}A = |A|I_3 = -\frac{1}{7}I_3$.

2. [10 pts.] Solve $\begin{vmatrix} 1 & 0 & 0 \\ 2 & x & 1 \\ 1 & 3 & x-5 \end{vmatrix} = \begin{vmatrix} x & -1 \\ 2 & x \end{vmatrix}$, for x .

We have $\begin{vmatrix} 1 & 0 & 0 \\ 2 & x & 1 \\ 1 & 3 & x-5 \end{vmatrix} = x(x-5) - 3$ and $\begin{vmatrix} x & -1 \\ 2 & x \end{vmatrix} = x^2 + 2$.

$$x(x-5) - 3 = x^2 + 2 \Leftrightarrow x^2 - 5x - 3 = x^2 + 2 \Leftrightarrow 5x = -5 \Leftrightarrow x = -1.$$

3. [10 pts.] Let $U = (1, 1, 2, -3, 1)$. Find all scalars k such that $\|kU\| = 4$.

We have $\|kU\| = |k|\|(1, 1, 2, -3, 1)\| = |k|\sqrt{1+1+4+9+1} = |k|\sqrt{16} = 4|k|$. Therefore, $\|kU\| = 4|k| = 4 \Leftrightarrow k = \pm 1$.

4. [10 pts.] Let X and Y be vectors in $\mathbb{R}^n$ such that $\|X\| = \|Y\| = 2$. Show that:

$$\|X + Y\|^2 + \|X - Y\|^2 = 16.$$

We have $\|X + Y\|^2 + \|X - Y\|^2 = (X + Y) \cdot (X + Y) + (X - Y) \cdot (X - Y) = \|X\|^2 + 2X \cdot Y + \|Y\|^2 + \|X\|^2 - 2X \cdot Y + \|Y\|^2 = 2\|X\|^2 + 2\|Y\|^2 = 2(4) + 2(4) = 16$.

5. [10 pts.] Let A be an $m \times n$ matrix. Show that the solution set of the homogenous linear system $AX = O$ in n unknowns is a subspace of $\mathbb{R}^n$.

Let W be the solution set of the system. The set W is not empty because it contains at least the trivial solution $X = O$.

To show that W is a subspace of $\mathbb{R}^n$, we must show that it is closed under addition and scalar multiplication. To do this, let X_1 and X_2 be vectors in W . Since these vectors are solutions of $AX = O$, we have $AX_1 = O$ and $AX_2 = O$. It follows that $A(X_1 + X_2) = AX_1 + AX_2 = O + O = O$ so W is closed under addition. Similarly, if k is any scalar then $A(kX_1) = kAX_1 = kO = O$ so W is also closed under scalar multiplication. Hence, W is a subspace of $\mathbb{R}^n$.

6. [10 pts.] Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $X = (x_1, x_2)$ and $Y = (y_1, y_2)$ in V and $k \in \mathbb{R}$:

$$X + Y = (x_1 + y_1 - 2, x_2 + y_2 - 2), \quad kX = (kx_1, kx_2).$$

Show that V is not a vector space under the given operations.

Since $0X = (0, 0) \neq O = (2, 2)$, therefore V is not a vector space under the given operations.

7. [10 pts.] Let $U_1 = (1, 2, 2, -1)$, $U_2 = (4, 9, 9, -4)$ be vectors in $\mathbb{R}^4$. Determine whether the vector $U = (5, 8, 9, -5)$ is in $\text{span}\{U_1, U_2\}$.

Form $c_1(1, 2, 2, -1) + c_2(4, 9, 9, -4) = (5, 8, 9, -5)$. This leads to a linear system whose

augmented matrix is:
$$\left[\begin{array}{cc|c} 1 & 4 & 5 \\ 2 & 9 & 8 \\ 2 & 9 & 9 \\ -1 & -4 & -5 \end{array} \right] \sim \left[\begin{array}{cc|c} 1 & 4 & 5 \\ 0 & 1 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array} \right].$$

It is clear that the linear system has no solution. Thus, U does not belong to $\text{span}\{U_1, U_2\}$.

8. [10 pts.] Determine whether the vectors $U_1 = (1, 2, 0)$, $U_2 = (2, 3, 0)$, and $U_3 = (4, 3, 1)$ span $\mathbb{R}^3$.

Since $\begin{vmatrix} 1 & 2 & 4 \\ 2 & 3 & 3 \\ 0 & 0 & 1 \end{vmatrix} = -1$, we conclude that the given vectors span $\mathbb{R}^3$ (by Theorem 2.3.8).

9. [$5 \times 4 = 20$ pts.] In each of the following questions, select the correct answer.

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- (C) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (4, 5, 7)\}$.
- (D) $\boxed{\{(1, 2, -1), (1, 0, 0), (1, 2, 0)\}}$.

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- (D) $-X = 1$.